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Problem Solving Protocol

A novel circRNA-miRNA association prediction model based on structural deep neural network embedding

Lu-Xiang Guo, Zhu-Hong You, Lei Wang, Chang-Qing Yu, Bo-Wei Zhao, Zhong-Hao Ren and Jie Pan
Corresponding authors: Zhu-Hong You. E-mail: zhuhongyou@gxas.cn; Lei Wang. E-mail: leiwang@gxas.cn

Abstract

A large amount of clinical evidence began to mount, showing that circular ribonucleic acids (RNAs; circRNAs) perform a very important function in complex diseases by participating in transcription and translation regulation of microRNA (miRNA) target genes. However, with strict high-throughput techniques based on traditional biological experiments and the conditions and environment, the association between circRNA and miRNA can be discovered to be labor-intensive, expensive, time-consuming, and inefficient. In this paper, we proposed a novel computational model based on Word2vec, Structural Deep Network Embedding (SDNE), Convolutional Neural Network and Deep Neural Network, which predicts the potential circRNA-miRNA associations, called Word2vec, SDNE, Convolutional Neural Network and Deep Neural Network (WSCD). Specifically, the WSCD model extracts attribute feature and behaviour feature by word embedding and graph embedding algorithm, respectively, and ultimately feed them into a feature fusion model constructed by combining Convolutional Neural Network and Deep Neural Network to deduce potential circRNA-miRNA interactions. The proposed method is proved on dataset and obtained a prediction accuracy and an area under the receiver operating characteristic curve of 81.61% and 0.8898, respectively, which is shown to have much higher accuracy than the state-of-the-art models and classifier models in prediction. In addition, 23 miRNA-related circular RNAs (circRNAs) from the top 30 were confirmed in relevant experiences. In these works, all results represent that WSCD would be a helpful supplementary reliable method for predicting potential miRNA-circRNA associations compared to wet laboratory experiments.

Keywords: circRNA-miRNA interaction, circRNA, miRNA, Structural Deep Network Embedding, deep neural network

Introduction

As a novel class of endogenous non-coding ribonucleic acid (RNA), circular RNAs (circRNAs) weigh a lot in the mammalian brain [1, 2]. Compared with the ribonucleic acid (RNA) formed by linear splicing, the circRNA does not have a 5′-terminal cap and 3′-terminal poly (A) tail, which special structure makes it more conservative and stable than linear RNA [3–5]. CircRNA was not known until 1976, when Kolakofsky and Sanger discovered plant infection virus and Sendai virus [6, 7]. Recently, circRNA has been successfully used for acting as microRNA (miRNA) sponge [8, 9]. Meanwhile, circRNA can translate proteins [10], regulate target gene expression [11], combine with related proteins [12], etc. More and more studies based on massive data analysis have shown that circRNA has essential biological functions and is an indispensable part of cells.

Along with the deepening understanding of a great quantity of circRNAs function, increasing evidence indicates that circRNAs have been explained to have complex associations with miRNAs [13–17]. For instance, a study by Anastas and Moon [18] revealed that circRNA cir-ITCH can be used as a sponge miR-7, miR-17 and miR-214 to inhibit the WNT pathway. Wu *et al.* [19] Confirmed that lifted levels of circRNA circTADA2A are associated with miR-203a-3p and have promoting effects on the progression of development of osteosarcoma. The myocardial ischemic injury model studied by mouse ischaemia–reperfusion by Li *et al.* [20] verified that circNCX1 is lifted in the model and is binding with miR-133a-3p, which can result in promoting cardiomyocyte apoptosis and growing expression of proapoptotic gene cell death-inducing protein. Therefore, exploring the potential correlations between circRNA and miRNA can contribute to biological scientists to diagnose complex pathogenesis of disease in the treatment of clinical diseases.

Although traditional biological experiment methods have discovered accurate results, which can excellently explore the relationship between circRNAs and miRNAs, it is still a time-consuming and labour-intensive assignment to identify the complicated relationship between them with some wet-lab experimental methods [21–24]. With more and more data on circRNAs accumulated, the increasing number of experimentally supported circRNA-miRNA association data has been increasingly discovered. The accumulation of these data offers a well-grounded foundation for prediction of circRNA-miRNA

Lu-Xiang Guo is a master's degree candidate at Xijing University. His research interests include bioinformatics, machine learning and complex network.
Zhu-Hong You is a professor at Northwestern Polytechnical University. His research interests include neural networks, intelligent information processing, sparse representation and its applications in bioinformatics.
Lei Wang is a professor at Guangxi Academy of Sciences. His research interests include data mining, machine learning, deep learning, computational biology and bioinformatics.
Chang-Qing Yu is a professor at Xijing University. His research interests include neural networks, data analysis and Internet of Things.
Bo-Wei Zhao is a PhD candidate at the University of Chinese Academy of Sciences. His research interests include machine learning, complex networks and graph neural network.
Zhong-Hao Ren is a master's degree candidate at Xijing University. His research interests include data mining, machine learning, deep learning and bioinformatics.
Jie Pan is a PhD candidate at Northwest University. His research interests include data analysis, pattern recognition, deep learning and computational biology.
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interactions using computer algorithm models. For example, Lan *et al.* [25] put forward a method of computing for NECMA, which constructs the heterogeneous network from multiple sources including circRNA GIP kernel similarity network, miRNA GIP kernel similarity network and the circRNA-miRNA association network, and accurately predicts the associations between circRNAs and miRNAs through using inner product and neighbourhood regularization logic matrix decomposition. The 10-fold cross-validation experiment results show that the model achieves the area under the receiver operating characteristic (ROC) curve (AUC) of 0.8264. Moreover, the top 10 circRNAs predicted by this model and these associations were confirmed by manually retrieving related literature and databases. Qian *et al.* [26] designed a computational framework called CMIVGSD based on graph variational auto-encoders and singular value decomposition to predict the interaction between circRNA and miRNA. The follow-up 5-fold cross-validation was performed. Experimental results show that the CMIVGSD achieved a prediction AUC of 0.8804. Besides that, six of the top ten circRNA-miRNA interactions (CMIs) predicted by the CMIVGSD model have been verified in PubMed.

Although the above existing prediction models have their encouraging performance. In addition, they play a significant role in accelerating the progress of circRNA-miRNA association identification. However, there is no denying that they still suffer from certain restrictions: (I) The current methods are mostly verified by the comparability information of circRNA and miRNA and do not employ complete related biological information included in the circRNA sequences to promote the analysis accuracy. (II) The number of proven circRNA-miRNA interactions is restricted, which cannot utilize the model adequately extract feature information, so the prediction model cannot be effectively trained.

To solve the above shortcomings, in this paper, as illustrated in Figure 1, we propose a new computational algorithm based on behaviour and attribute feature, to predict circRNA-miRNA associations. To be more precisely speaking, three main steps constitute the framework of the model: firstly, we extracted behaviour feature learned by graph embedding and attribute feature learned by word embedding, and then integrated the above two types of information to represent each particular node, and then experimentally verified interaction samples and an equal number of unlabelled pairs are utilized to construct positive samples and random selection of negative samples, respectively, and ultimately fed them into a feature fusion model constructed based on CNN and DNN to infer underlying circRNA-miRNA associations.

Materials and methods

Dataset

In this study, we predict experiments based on the circRNA-miRNA dataset derived from wet laboratory experiments, which are downloaded from CircBank [27] database (http://www.circbank.cn/) and miRbase [28, 29] database (https://www.mirbase.org/). CircBank schemed a comprehensive database that supports the CMIs, which includes more than 140,000 human annotated circRNAs from a variety of sources.

After processing redundant data, we built a dataset that collects high-quality 20,208 CMIs from 3569 circRNAs and 1152 miRNAs in CircBank. The number of independent miRNAs, circRNAs and miRNA-circRNA interactions in dataset (R) can be seen in Table 1. Therefore, we divide the dataset as follows in the study:

$$R = R^{+} \cup R^{-}. \tag{1}$$

Here R^{+} indicates the positive sample dataset, which clarifies supported 20,208 miRNA-related circRNAs in the dataset. R^{-} represents an unlabelled sample set, that is circRNA-miRNA associations that are possible not verified by experiments, and its number is $3569 \times 1152 - 20{,}208 = 4{,}091{,}280$. According to the above explanation, we structured the adjacency matrix A based on our constructed dataset with the dimension of 3569×1152. When circRNA $c(i)$ and miRNA $m(j)$ associations are verified, the element $A(i, j)$ of A is given a value of 1; otherwise, it is set to 0.

Word embedding

Word embedding [30, 31] is a way of word processing in NLP (natural language processing) [32] language model. They are vectors for representing the meaning of words through algorithms, which can describe sequence information [32–34]. However, word embeddings did not become the foundations of modern deep-NLP until Mikolov, K. Chen, *et al.* [35] proposed two very simple log-linear models. Of course, there have been plenty of suggestions to perfect Word2vec, but after 9 years it continues to be used regularly as a solid baseline and the default source of embeddings. Specifically, Word2vec has made significant progress in computational biology knowledge. In this study, considering the similarity between biological sequence and text, and the accumulated knowledge in the field of natural language processing, Word2vec was used to sequence analysis to numerically describe the circRNA and miRNA. Word2vec falls in two flavours: CBOW and Skip-Gram.

In this work, the context vectors of CBOW [36] are summed and used as target predictions. The maximized objective function is as follows:

$$\frac{1}{T}\sum_{t=1}^{T}\log p\left(\omega_t \middle| \sum_{-c\le j\le c, j\ne 0} \omega_{t+j}\right). \tag{2}$$

The maximized objective function value in Formula 2 is the output value of CBOW. Let there be a corpus, T is the size of the corpus and $w_1, w_2, \ldots, w_T$ stands for a sequence of words. Parameter c signifies the window, where we use c words on the right and left of the target.

In the study, we applied the recently established gensim python package based on experimental validation as the language pre-training model to measure performance. When there are two parameters, '**min**_count' and 'size', the values of the corresponding parameters are expressed as 1 and 64, respectively, else parameters are expressed as default. Finally, a 64-dimensional vector is taken as each node representation.

Graph embedding

In recent years, Graph embedding [37, 38], which aims to convert the graph data into low-dimensional space, has drawn sustained concern. Graph embedding can well prepare for feature extraction by describing sequence information of circRNA and miRNA. In this paper, Structural Deep Network Embedding (SDNE) [39, 40] is a semi-supervised deep autoencoder model for deep learning that captures non-linearity in network information and structure.

In the unsupervised part, the model about deep autoencoder is structured by SDNE, which learns the second-order similarity to preserve vertex representations. For capturing the second-order proximity, we reconstructed the $|V|$-dimensional vertex adjacent matrix representation that tries to minimize

$$L_{2nd} = \sum_{i=1}^{|v|} \left\| \left(r_{v_i}^{(0)} - \hat{r}_{v_i}^{(0)}\right) e\ b_i \right\|. \tag{3}$$

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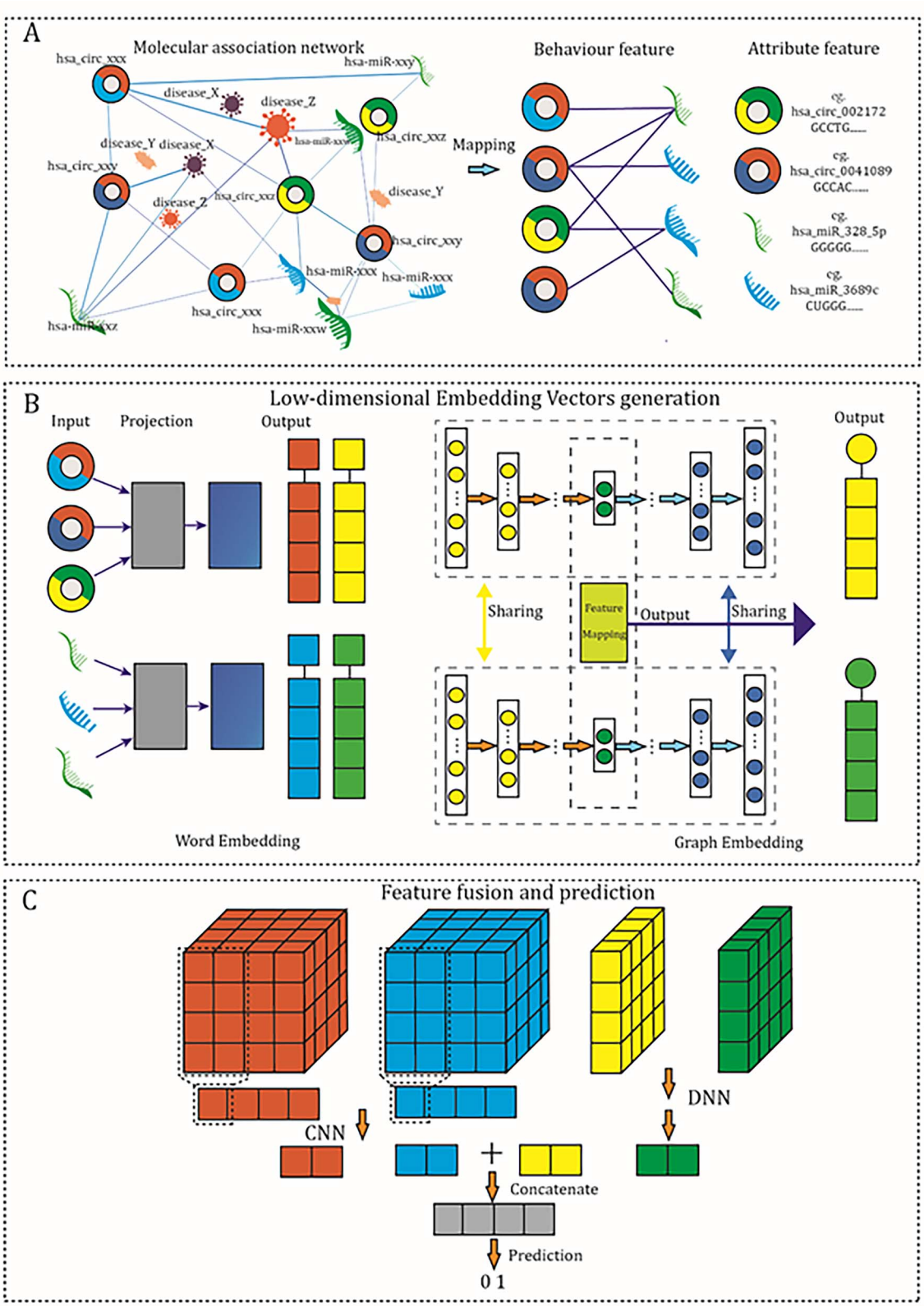

Figure 1. The flowchart of WSCD model. **A** Construction of Molecular Association Network. **B** The extract workflow of attribute feature and behavior feature. **C** The workflow of feature fusion and prediction.

Here the input representation and the reconstructed representation are $r_{v_i}^{(0)} = S_{i:}$ and $\hat{r}_{v_i}^{(0)}$, respectively. To penalize non-zero elements of s' with construction error, we set b_i as a weight vector.

In the unsupervised part, the first-order proximity is imported by SDNE via penalizing the distance, which is between connected vertices conserved in the embedding space. This loss function of objective is defined as

$$L_{1st} = \sum_{i=1,j=1}^{|v|} S_{ij} \left\| \left(r_{v_i}^{(K)} - r_{v_j}^{(K)} \right) \right\|_2^2, \quad (4)$$

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Table 1. Detailed information of CMIs used in WSCD

Dataset	circRNA	miRNA	interactions
CircBank	3569	1152	20,208

where K stands for the number of hidden layers and $r_{v_i}^{(K)}$ represents the Kth layer representation of vertex v_i. In all, the final loss function is the joint objective function that is minimized by SDNE:

$$L = L_{2nd} + \alpha L_{1st} + \upsilon L_{reg}. \tag{5}$$

Here L_{reg} is a regularization term that prevents overfitting. After solving the minimization of (5), for vertex v_i, the Kth layer representation $r_{v_i}^{(K)}$ is taken as its representation r_{v_i}. When the minimization of (5) is resolved, vertex v_i for the Kth layer representation $r_{v_i}^{(K)}$ is regarded as its representation r_{v_i}.

In this paper, we also adopted the established open-NE library to perform the graph embedding task. When there are two parameters, 'order' and 'representation – size', the values of the corresponding parameters are expressed as 1 and 64, respectively, else parameters are expressed as default. The successive k nucleotides are deemed as one word and we set k as 1. Finally, the size of the matrix is the 'length of sequence* 64' which is taken as the size of each 'word' representation.

A deep learning fusion model combining CNN and DNN

Convolution neural networks (CNNs) [41, 42] and Deep Neural Networks (DNNs) [43, 44] obtain high performance in feature extraction, respectively, and are also widely used in biology knowledge. However, unilateral feature extraction is not comprehensive enough. In this paper, we proposed a deep learning fusion model [45] based on ensemble CNN and DNN as shown in Figure 2.

In the process of feature extraction [41, 45], it can be seen that the first step processes of CNN from the input layer to the convolutional layer, pooling layer, dense layer, and first step processes of DNN from the input layer to the hidden layer, the second step is by a fully connected layer will extract behavior and attribute feature to integrate the features.

In the feature fusion problem, as for the training set $D = \{X, X', Y\}$ and has N samples here N represents the training sample, and the number X is the input data; the label of the X is Y; and we denote the forward propagation process of feature extraction with iteration t as

$$f\left(x, x'\right) = f_{L-1}\left(f_{c,L-2}\left(\ldots f_{c,1}\left(x, \theta_{c,1}^{t}\right), \theta_{c,L-2}^{t}\right),\right.$$
$$\left. f_{d,L-2}\left(\ldots f_{d,1}\left(x', \theta_{d,1}^{t}\right), \theta_{d,L-2}^{t}\right), \theta_{L-1}^{t}\right), \tag{6}$$

$$O^{L} = \frac{\exp f_{L}\left(x, x', \theta_{L}^{t}, c\right)}{\sum_{j=1}^{c} \exp f_{L}\left(x, x', \theta_{L}^{t}, j\right)}. \tag{7}$$

The fusion model learnable parameters are represented as $\theta_{c,1}^{t}, \theta_{c,2}^{t}, \ldots, \theta_{c,L-1}^{t}, \theta_{d,1}^{t}, \theta_{d,2}^{t}, \ldots, \theta_{d,L-1}^{t}$, respectively. The parameters $f_{c,L-1}$ and f_{d}^{L-1} are represented by each network layer operations of CNN and DNN, and x is expressed as the input samples of CNN and DNN. On input x and x', $f_L(x, x', \theta_L^t)$ is the output parameter of L network layer. The parameter O^L is represented by the result of the classification at output layer.

Results and discussion

Evaluation indicators

In the experiment, every potential verified circRNA–miRNA pair in the dataset (R) was used, in which experimental proven interaction samples were constructed as the positive sample set, and an equal number of unlabelled pairs were identified as negative samples. To decrease overfitting in the experimental information to a certain extent, a reliable 5-fold CV technique is applied to assess the prediction performance of our proposed approach. Finally, we carried out 50 experiments and took the average as the final result. The MCC (Matthews's correlation coefficient), Spec. (specificity), Prec. (Precision), Sen. (Sensitivity) and Acc. (Accuracy) are applied separately as the common evaluation indicators to evaluate the model performance and fairness of the results. These indicators are defined as follows:

$$\text{Acc.} = \frac{TP + TN}{TP + TN + FP + FN}, \tag{8}$$

$$\text{Sen.} = \frac{TP}{TP + FN}, \tag{9}$$

$$\mathbf{Pr}\,e. = \frac{TP}{TP + FP}, \tag{10}$$

$$\text{Spec.} = \frac{TN}{TN + FP}, \tag{11}$$

$$\text{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)\,(TP + FN)\,(TN + FP)\,(TN + FN)}}. \tag{12}$$

Here TP (true positive) and TN (true negative) stand for the number of miRNA-associated circRNAs pairs that were precisely predicted and imprecisely predicted, FP (false positive), and FN (false negative) stand for the quantity of CMIs pairs that were precisely predicted and imprecisely predicted, respectively. Besides, the AUC (Area Under ROC), AUPR (Area Under Precision-Recall (PR)) and ROC (Receiver-Operating-Characteristic) are widely employed to evaluate the classifier's performance. The ROC and Precision Recall (PR) are used to plot for visualization.

Model performance evaluation

We first used evaluation indicators to assess the prediction performance of our proposed model. The 5-fold CV was performed to generate the results of the above evaluation indicators on our constructed dataset. To correctly estimate the performance evaluation of the model, we conducted this work using the 5-fold CV experiment. The prediction mean results are demonstrated in boldface in Table 2. As the table can be seen, WSCD (**W**ord2vec, SDNE, **C**onvolutional Neural Network and **D**eep Neural Network) obtained 81.61% mean prediction accuracy using 5-fold CV experiments, and the results of each fold experiment were 82.46%, 82.81%, 80.90%, 80.64% and 81.26%, respectively, while its Standard Deviation (SD) was only 0.97%.

We came up with a computational model for WSCD, which achieves the AUC reflecting the comprehensive performance of the model, and the AUC accurately achieved 0.0070 SD with 0.8898 average prediction accuracy. Other evaluation parameters including Sen., Prec., Spec. and MCC, WSCD achieved SDs were 1.11%, 0.91%, 0.92% and 1.92%, and their excellent proportions of 81.91%, 81.43%, 81.32% and 63.23%, respectively. Figure 3 presents the AUPR of the proposed method and the ROC curves generated by the results of WSCD on the dataset using 5-fold CV. From these above-mentioned measures, it can be demonstrated that WSCD

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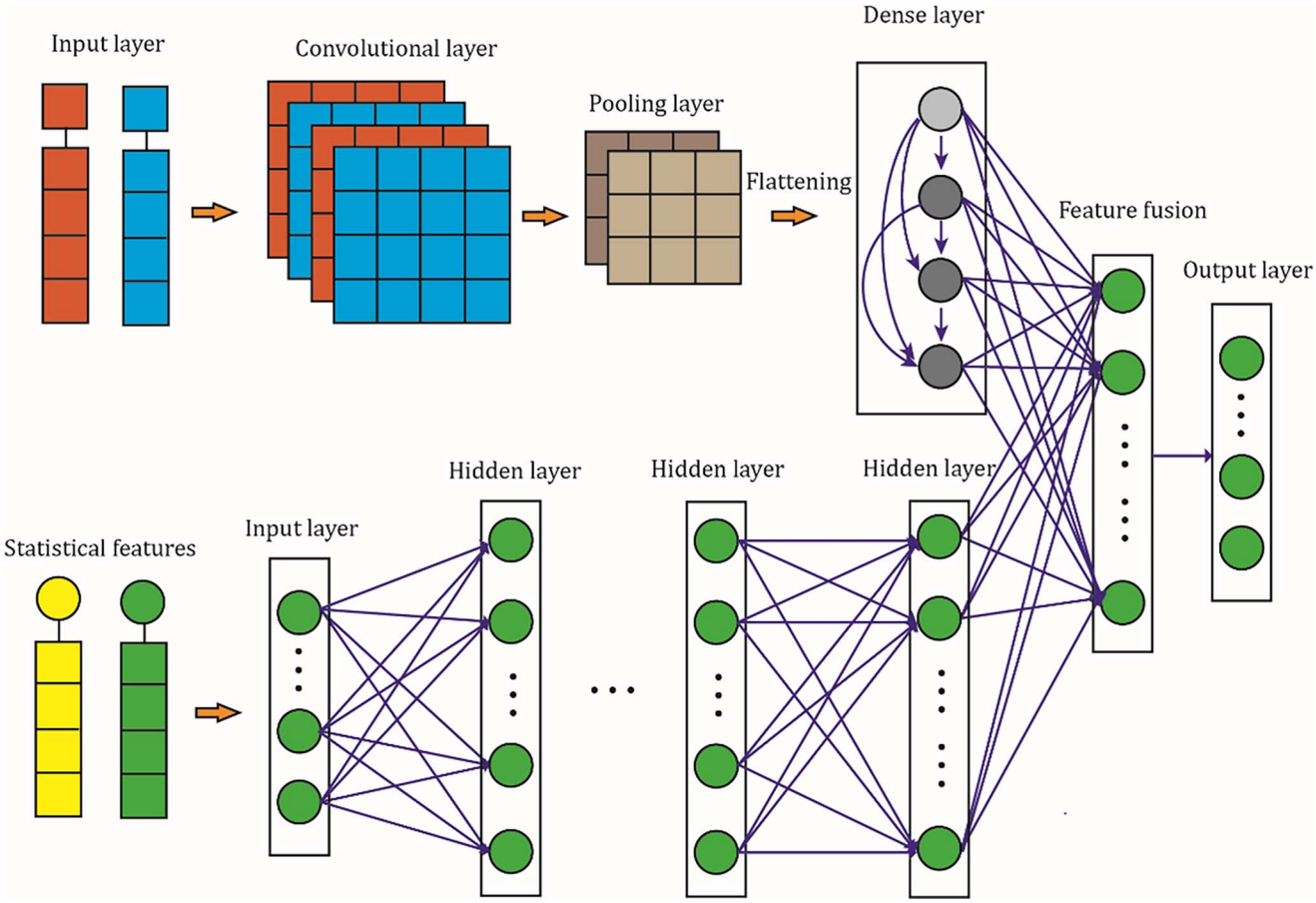

Figure 2. Feature extraction process of WSCD model.

Table 2. Results of 5-fold CV obtained by WSCD. The prediction mean results and SD are demonstrated in boldface.

5-fold	Acc. (%)	Sen. (%)	Spec. (%)	Prec. (%)	MCC (%)	AUC
fold1	82.46	82.93	81.99	82.16	64.92	0.8988
fold2	82.81	83.28	82.34	82.5	65.61	0.8944
fold3	80.90	81.30	80.50	80.66	61.80	0.8837
fold4	80.64	81.04	80.25	80.4	61.29	0.8823
fold5	81.26	80.99	81.53	81.43	62.53	0.8896
Average	**81.61**	**81.91**	**81.32**	**81.43**	**63.23**	**0.8898**
SD	**0.97**	**1.11**	**0.92**	**0.91**	**1.92**	**0.0070**

has high performance and is effective and stable for the CMIs prediction.

We can see that the main computational cost of our proposed model is the feature extraction stage using CNN and graph embedding). Its time cost is $O(Kd)$ for each sample, where there are two parameters, K and d, the values of the corresponding parameters are expressed as the dimension and the number of features in each sample, respectively. As such, the total complexity of WSCD is $O(TNKd)$, where T is the number of iterations and N is the number of samples. In the work, we used PyCharm Community Edition 2021.1×64 to run the programme and the server was configured with Intel(R) Core (TM) i7-12700H CPU and 16 GB RAM. As we can see, each fold dataset in 5-fold CV takes the average processing time is 150.588, 150.562, 229.927, 243.978 and 229.978 s, respectively.

Comparison among different classifiers

In the feature fusion model, we use the DNN and CNN to construct the classifier inferring potential interactions. To ensure the influence of the performance of the classifier, we carried out a comparative experiment of different classifiers in the study. In particular, we kept Word embedding and Graph embedding unchanged, and merely substitute the feature fusion model with classifiers containing GBDT (Gradient Boosting Decision Tree) [46–48], KNN (K-Nearest Neighbour) [49], LR (Logistic Regression) [50], RoF (Rotation Forest) [51] and SVM (support vector machine) algorithm [52] for experiments. Evaluations of these classifier models using the same attribute and behaviour feature on training set data are summarized in Table 3, where the bold values represent the maximum values obtained by our model. Using the same dataset in 5-fold CV, we trained above different classifier models,

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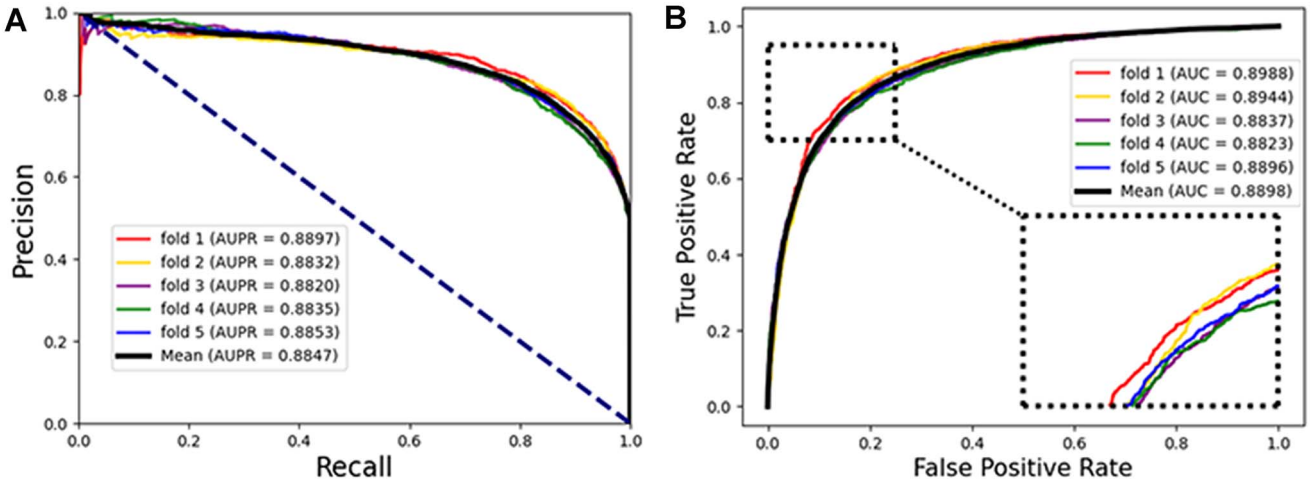

Figure 3. Performance evaluation of our proposed method. **A** AUPR generated by the results of WSCD on the dataset using 5-fold CV. **B** ROC curves generated by the results of WSCD on the dataset using 5-fold CV.

Table 3. Comparison of evaluation indicators of different classifiers of this pape. The bold values represent the maximum values.

Indicator	GBDT	KNN	LR	RF	SVM	Our
Acc. (%)	72.12 ± 0.59	67.31 ± 0.69	72.24 ± 0.79	67.25 ± 1.22	70.64 ± 0.83	**81.61 ± 0.97**
Sen. (%)	67.42 ± 1.25	67.28 ± 1.40	66.64 ± 1.02	69.44 ± 1.90	66.27 ± 1.25	**81.91 ± 1.11**
Spec. (%)	76.83 ± 0.84	67.35 ± 0.89	77.84 ± 0.87	65.06 ± 1.11	75.01 ± 0.51	**81.32 ± 0.92**
Prec. (%)	74.42 ± 0.63	67.32 ± 0.61	75.05 ± 0.89	66.52 ± 1.05	72.61 ± 0.73	**81.43 ± 0.91**
MCC (%)	44.45 ± 1.15	34.63 ± 1.37	44.77 ± 1.58	34.54 ± 2.45	41.43 ± 1.65	**63.23 ± 1.92**
AUC	0.5446 ± 0.0142	0.5361 ± 0.0135	0.4966 ± 0.0118	0.4804 ± 0.0173	0.5743 ± 0.0009	**0.8898 ± 0.0070**

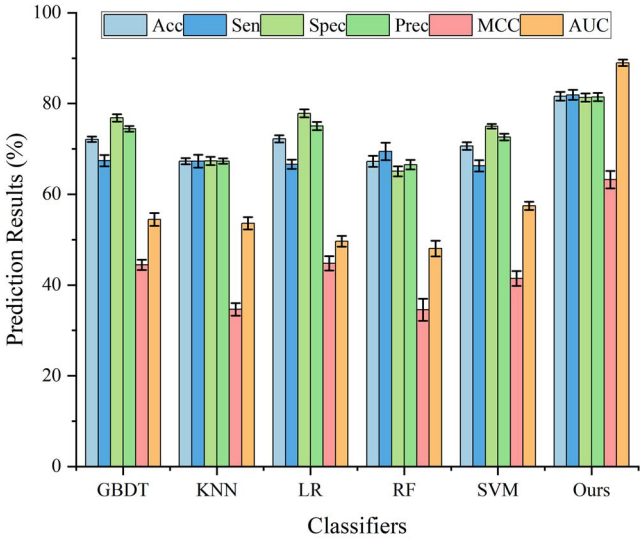

Figure 4. Comparison of outcomes gained by different classifier models.

summarized in Supplementary Tables S1–S5, detailedly. The best performing configurations of the fine-tuned classifiers are appended below the above supplementary tables. However, we present the same results in the form of the histogram in Figure 4, for the convenience of intuitive comparison with such classifiers. As can be seen in Figure 4, our proposed classifier model obtains the best result. These overall results represent performance that indicates the feature fusion model based on CNN and DNN can effectively predict the CMIs.

Comparison of performance of different strategies

In this chapter, we compared the proposed model with attribute and behaviour feature fusion with several existing state-of-the-art models. To be fair, we use the method of the two kinds of features to construct the representation vectors and train the computational model, respectively, while the rest of the parts of the model remain the same. There are three strategies: 'A', 'B' and 'A + B'. The meaning of the corresponding parameters is expressed as attribute-based, behaviour-based and a combination of both methods, respectively, and the results can be seen in Figure 5. The ACC (accuracy) of the same outcomes is presented as a histogram in Figure 5. Figure 5 represents the results of 5-fold CV generated on the dataset by three models, which shows 'A + B' incorporating the two kinds of feature achieves a higher prediction accuracy of 81.61% than others. Compared to Figure 3, Figure 6 also shows the advantages of 'A + B' through the ROC curves.

Comparison with k-mer model

To verify the influence of WSCD using attribute and behaviour feature as model attributes to enhance this model performance, we compared it with the feature vector generated by k-mer frequencies and probability. For the sake of fairness, we used the method of the feature vector extracted by k-mer frequencies and probability to replace the word embedding and the graph embedding during the experience process, and leaving the rest parts of the model unvaried. Using the same dataset in 5-fold CV, we trained the model with k-mer frequencies and probability, summarized in Table 4 with the comparison results, detailedly. As can be seen in Table 4, WSCD obtains the optimum result, and

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Table 4. Outcomes of 5-fold CV acquired by K-mer model

5-fold	Acc. (%)	Sen. (%)	Spec. (%)	Prec. (%)	MCC (%)	AUC
fold1	74.59	69.77	79.42	77.22	49.41	0.8087
fold2	74.42	70.56	78.28	76.46	48.98	0.8113
fold3	74.42	68.88	79.96	77.46	49.14	0.8062
fold4	74.03	67.97	80.10	77.35	48.43	0.8010
fold5	75.62	72.77	78.47	77.17	51.32	0.8082
Average	74.62	69.99	79.25	77.13	49.46	0.8071
SD	0.60	1.83	0.84	0.39	1.10	0.0038

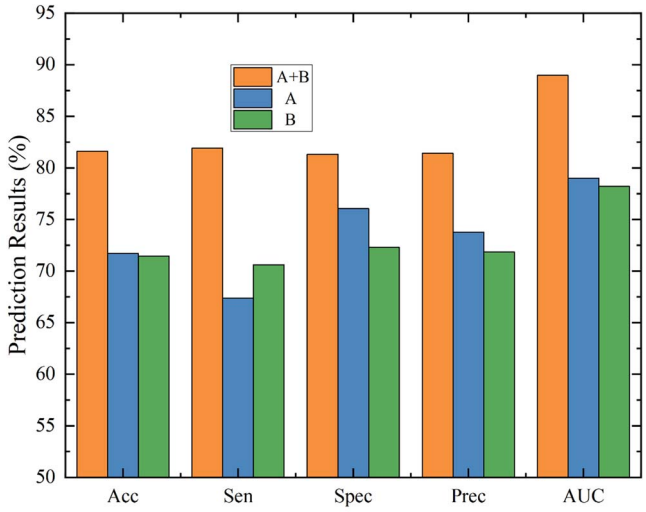

Figure 5. Prediction performance of different strategies. 'A', 'B' and 'A + B' represent attribute-based, behaviour-based and the combination of above both methods, respectively.

its prediction Acc., Sen., Prec., Spec., MCC and AUC are higher than the model with k-Mer frequencies and probability by 6.99%, 11.92%, 4.3%, 2.07%, 13.77% and 8.27%, respectively. This result shows that the Word2vec and SDNE methods used by the WSCD model can effectively extract attribute and behaviour features, which helps improve the model and achieve the most competitive performance.

Contrast and other pre-existing models

With the continuous accumulation in computational algorithms, more and more machine learning models have been introduced to tackle the relationship prediction between circRNA and miRNA. Since there's a fair consideration here, we compare CMIVGSD [26] with several of the previously proposed models, which use the same parameter of the AUC score. The previously proposed models include NECMA [25] and CMIVGSD.

Such sufficient data information is required by the above methods. However, the limited proverbial CMIs data and the disequilibrium of positive and negative samples affect their performance. NECMA and CMIVGSD use methods based on heterogeneous network to represent features between circRNA and miRNA. To overcome the heterogeneous networks sparsity caused by algorithm iterations, network attributes are enhanced to improve performance. As can be concluded from the comparison of the results, the AUPR value of WSCD is 0.8898 which is higher than NECMA (0.0634) and CMIVGSD (0.0094), and thereby our model is the most competitive.

Case studies

To further evaluate the assessment of the ability of WSCD, in this paper, we carried out a case study. Specifically, we used all experimentally known miRNA-related circRNAs pairs to train WSCD, and predicted all unknown circRNA-miRNA associations. In addition, the higher scores of miRNA-related circRNAs pairs are selected to confirm whether they were verified by relevant literature or experiments in biology. In all the predicted results, 23 of the top 30 miRNA-related circRNAs pairs selected were confirmed, and details of these are seen in Table 5. The collection of such data offers a well-grounded demonstration for the effective prediction of circRNA–miRNA interactions using our proposed computer algorithm models, making up for the deficiency of the biological experiments.

Results and discussion

Accumulating the development of high-throughput technology experiments has shown that richness of genomic data not only helps to comprehend intricate disease pathogenesis but also provides effective help for preventing and diagnosing diseases and precision medicine. The relationship between circRNA and miRNA undoubtedly plays a crucial part in helping to understand human pathological processes, decipher the cellular behaviour and identify biomarkers for diagnosis and treatment. It can be seen that the proposed WSCD has competitive computational advantages, analysing large-scale genetic data and removing false positive signals, effectively, when compared with the pre-existing wet experiments methods. Besides, this model can obtain basic scientific research outcomes without any laboratory equipment, thanks to the rapid development of computers, especially the computing and storage ability, reduce the man-made interference of experiments and promote the advances of experiments, so as to make more reliable inferences of biological discovery.

In this paper, we propose a novel deep learning framework model, WSCD, to infer CMIs, and this model is supported by NLP algorithms, graph embedding and neural networks algorithms. The WSCD model uses NLP algorithms to depict circRNA and miRNA sequence data, which can greatly enrich the statement ability of descriptors. In the feature extracting period, NLP can enrich the expression ability and graph embedding low-dimensional ability, and they can effectually extract the essential features of miRNA-related circRNAs information, thereby perfecting the model prediction. In classification prediction, the neural network algorithms can continuously optimize searching for the global optimal solution and improve the prediction accuracy. Thus, we can achieve competitive results in the experiment by combining the above advantages. However, there were still some limitations to this model. As we all know, features extracted

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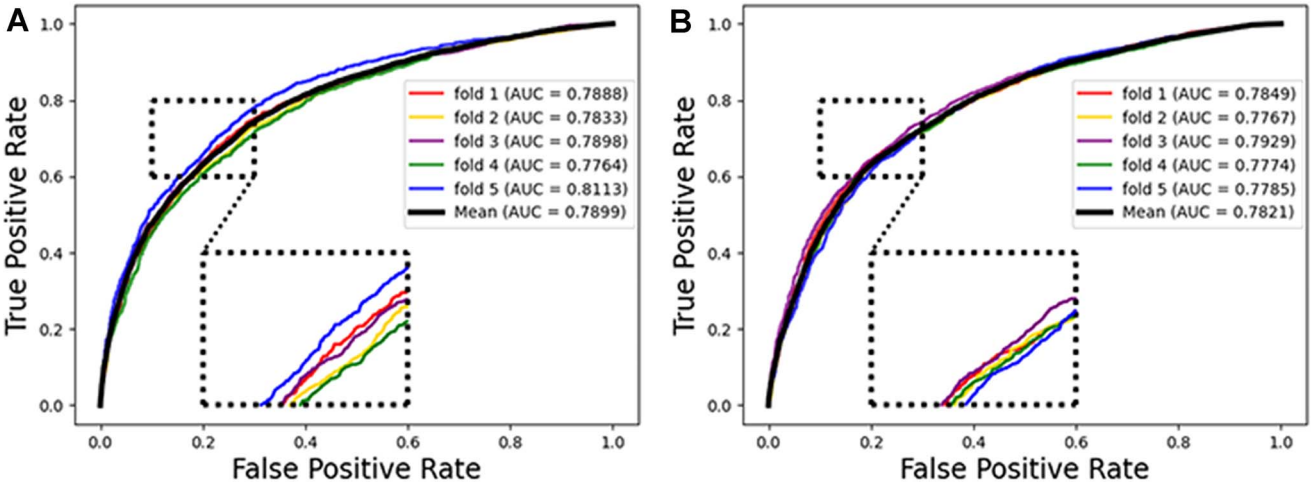

Figure 6. Performance evaluation of different strategies. **A** AUC is achieved by the proposed attribute-based method. **B** AUC is achieved by the proposed behavior-based method.

Table 5. Top 30 CMI pairs predicted by WSCD

Rank	circRNA	miRNA	Evidence	Rank	circRNA	miRNA	Evidence
1	hsa_circ_0044246	hsa-miR-4685-5p	Confirmed	16	hsa_circ_0052102	hsa-miR-6756-5p	Unconfirmed
2	hsa_circ_0044246	hsa-miR-4685-5p	Confirmed	17	hsa_circ_0010610	hsa-miR-4726-5p	Confirmed
3	hsa_circ_0044246	hsa-miR-762	Confirmed	18	hsa_circ_0010691	hsa-miR-6756-5p	Confirmed
4	hsa_circ_0049494	hsa-miR-762	Confirmed	19	hsa_circ_0081596	hsa-miR-4685-5p	Confirmed
5	hsa_circ_0009563	hsa-miR-661	Confirmed	20	hsa_circ_0009652	hsa-miR-6756-5p	Confirmed
6	hsa_circ_0049494	hsa-miR-4726-5p	Confirmed	21	hsa_circ_0055327	hsa-miR-4640-5p	Unconfirmed
7	hsa_circ_0082879	hsa-miR-1184	Confirmed	22	hsa_circ_0046337	hsa-miR-1184	Unconfirmed
8	hsa_circ_0040923	hsa-miR-877-3p	Confirmed	23	hsa_circ_0049494	hsa-miR-3620-5p	Unconfirmed
9	hsa_circ_0039075	hsa-miR-1184	Confirmed	24	hsa_circ_0016778	hsa-miR-5193	Unconfirmed
10	hsa_circ_0081596	hsa-miR-4763-3p	Confirmed	25	hsa_circ_0040863	hsa-miR-6756-5p	Confirmed
11	hsa_circ_0004566	hsa-miR-4726-5p	Unconfirmed	26	hsa_circ_0026097	hsa-miR-4726-5p	Confirmed
12	hsa_circ_0010610	hsa-miR-4685-5p	Confirmed	27	hsa_circ_0051748	hsa-miR-762	Confirmed
13	hsa_circ_0044512	hsa-miR-6756-5p	Confirmed	28	hsa_circ_0022613	hsa-miR-6756-5p	Confirmed
14	hsa_circ_0051748	hsa-miR-4763-3p	Confirmed	29	hsa_circ_0055327	hsa-miR-4763-3p	Confirmed
15	hsa_circ_0004566	hsa-miR-4763-3p	Unconfirmed	30	hsa_circ_0009563	hsa-miR-1184	Confirmed

by NLP algorithms cannot fully explain their biological significance, and two feature extraction methods can only deal with non-negative data. In addition, deep learning approaches generally achieved certain errors in prediction results because of the parameter settings.

In this work, the model solves the prediction problem by a deep learning fusion model combining CNN and DNN, where fusion model performs prediction tasks and combines miRNA and circRNA information using Word2vec and SDNE to generate attribute and behaviour feature and integrate these features. Excellent results have been obtained by WSCD on the dataset. To verify the contribution of attribute and behaviour feature, we sufficiently carried out comparative experiments, respectively. Furthermore, we confirmed 23 pairs from the top 30 pairs by checking and testifying the relevant literature. In all, WSCD is a precise model of circRNA–miRNA prediction and a useful tool offers credible candidates for biomedicine. Another line of the next study is to research large feature dimensions and a mass of training samples, to further enhance the ability of the model.

Competing interests

The authors declare that they have no competing interests.

Key Points

- Merging multiple biological information and incorporating attribute features and behaviour features to describe the circRNA-miRNA association comprehensively.
- We present a WSCD model, which is adopted to train the model through all CMIs to more accurately predict CMIs
- Through abundant comparative experiments, the experimental results and reliability are guaranteed, and theoretical guidance and direction are provided for subsequent experimental research.

Supplementary data

Supplementary data are available online at https://academic.oup.com/bib.

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Data availability

The data can be freely downloaded from https://github.com/look0012/WSCD.

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